EXP-3 : To encrypt pairs of letters using a keyword-based 5 x 5 matrix.

PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <string.h>
int main(void) {
char seed[] = "MONARCHYABCDEFGHIKLMNOPQRSTUVWXYZ";
char box[26] = "", text[] = "BALLOON", p[100];
int used[26] = {0}, n = 0, m = 0;
for (int i = 0; seed[i]; i++)
if (!used[seed[i] - 'A']++) box[n++] = seed[i];
for (int i = 0; text[i]; ) {
char a = text[i++], b = text[i];
if (!b || a == b) b = (a == 'X' ? 'Q' : 'X');
else i++;
p[m++] = a; p[m++] = b;
}
p[m] = 0;
printf("Prepared text: %s\nCiphertext: ", p);
for (int i = 0; i < m; i += 2) {
int a = (int)(strchr(box, p[i]) - box);
int b = (int)(strchr(box, p[i+1]) - box);
int r = a/5, c = a%5, s = b/5, d = b%5;
if (r == s) { c = (c+1)%5; d = (d+1)%5; }
else if (c == d) { r = (r+1)%5; s = (s+1)%5; }
else { int t = c; c = d; d = t; }
printf("%c%c", box[5*r+c], box[5*s+d]);
}
puts("");
return 0;
}
```

OUTPUT :

```
Prepared text: BALXLOON
Ciphertext: IBSUPMNA


...Program finished with exit code 0
Press ENTER to exit console.
```